SCIENCE AND TECHNOLOGY OF MATERIALS – TECHNOLOGY OF METALLIC MATERIALS

Steels Strengthening Excercise

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PROBLEM 1

Al-Cu is a ‘well-studied’ system for the study of precipitation hardening. For the Al 4 at.% Cu solid solution with the following parameters:
The lattice parameter of Al = 4.05 Å.
The shear modulus of Al = 26 GPa
Poisson’s ratio = 0.33.
The atomic volume change for Cu atom (ΔV) = 10 Å^3.
Taylor coefficient (α) = 0.5
Suppose that the supersaturated alloy is aged for 1000 h and the volume fraction of θ phase grows to 10%. The matrix still contains 2 at.% Cu.

CALCULATE:

A → The (shear) strength of supersaturated solid solution whose dislocation density is 10^{12} m^{-2} at room temperature.

B → The mean spacing between the precipitates. Assume that precipitates are spherical with the diameters of 100 nm, and that particles form a cubic array with the spacing L.

C → The (shear) strength of this equilibrated composite with precipitate θ with the dislocation density of 10^{12} m^{-2}. (hint: Add all possible contributions.)

D→ Perform the same calculation in (C) with the particle diameter of 200 nm.

PROBLEM 1 – SOLUTION A

The (shear) strength of supersaturated solid solution whose dislocation density is 10^{12} m^{-2} at room temperature.
Suppose that the supersaturated alloy is aged for 1000 h and the volume fraction of θ phase grows to 10%. The matrix still contains 2 at.% Cu.

(a) The shear strength of supersaturated solid solution can be obtained by summing the Taylor relation and the increment in yield strength by solid solution. First of all, let's calculate the magnitude of Burgers vector, which is $b = \frac{\sqrt{2}}{2} a = \frac{\sqrt{2}}{2} \times 4.05\dot{A} = 2.8638\dot{A}$. The total strength can be calculated as $\tau = \tau^{Taylor} + \tau^{Solid\ Solution}$.

$$\tau^{Tayor} = \alpha \mu b \sqrt{\rho} = 0.5 \times (26 \times 10^{3}\ MPa) \times (2.8638 \times 10^{-10}\ m) \times \sqrt{10^{12}\ m^{-2}} = 3.723\ MPa\ .$$

$$\tau^{Solid\ Solution} = \frac{\sqrt{3}(1+\nu)\mu \Delta V}{8\pi(1-\nu) b^{3}} \sqrt{x_s} = \frac{\sqrt{3}(1+0.33)(26 \times 10^{3}\ MPa)(10 \times 10^{-30} m^{3})}{8\pi(1-0.33)(2.8638 \times 10^{-10}\ m)^{3}} \sqrt{4 \times 10^{-2}}$$

$$= 302.9\ MPa$$

So, $\tau = 3.723 + 302.9 = 306.6\ MPa$.

PROBLEM 1 – SOLUTION B

The mean spacing between the precipitates.
Assume that precipitates are spherical with the diameters (d) of 100 nm, and that particles form a cubic array with the spacing L.

According to the given assumption, the mean spacing between the particles can be obtained as follows.

$$(volume fraction) = \frac{(\#\ of\ \ particle\ in\ the\ cubic,\ L^3) \times (the\ volume\ of\ \ the\ \ particle)}{(the\ volume\ of\ \ cubic,\ ,\ L^3)}$$

$$= \frac{1 \times \frac{4}{3}\pi r^3}{L^3} = 0.1$$

Thus,

$$L = \sqrt[3]{\frac{4}{3}\pi r^3 \times \frac{1}{0.1}} = \sqrt[3]{\frac{4}{3}\pi (50 \times 10^{-9}\ m)^3 \times \frac{1}{0.1}} = 173.7\ nm$$

PROBLEM 1 – SOLUTION C

The (shear) strength of this equilibrated composite with precipitate θ with the dislocation density of $10^{12}\ m^{-2}$.
(hint: Add all possible contributions.)

The total strength can be calculated by summing all possible contribution such as Taylor strengthening, solid solution strengthening and precipitation strengthening. So,

$$
\begin{aligned}
\tau &= \tau^{Taylor} + \tau^{Solid\ Solution} + \tau^{PPT} \\
&= \alpha \mu b \sqrt{\rho} + \frac{\sqrt{3}(1+\nu)\mu \Delta V}{8\pi(1-\nu)b^3}\sqrt{x_s} + \frac{\mu b}{L-d} \\
&= 3.723 + 214.2 + \frac{(26\times10^3)(2.8638\times10^{-10}\ m)}{(173.7-100)\times10^{-9}\ m}\ (MPa) \\
&= 319\ MPa
\end{aligned}
$$

PROBLEM 1 – SOLUTION D

Perform the same calculation in (C) with the particle diameter of 200 nm.

For the 200 nm diameter particle, the mean spacing is 347.3 nm. So,

$$\tau = \tau^{Taylor} + \tau^{Solid\ Solution} + \tau^{PPT}$$

$$= \alpha \mu b \sqrt{\rho} + \frac{\sqrt{3}(1+\nu)\mu \Delta V}{8\pi(1-\nu)b^3}\sqrt{x_s} + \frac{\mu b}{L-d}$$

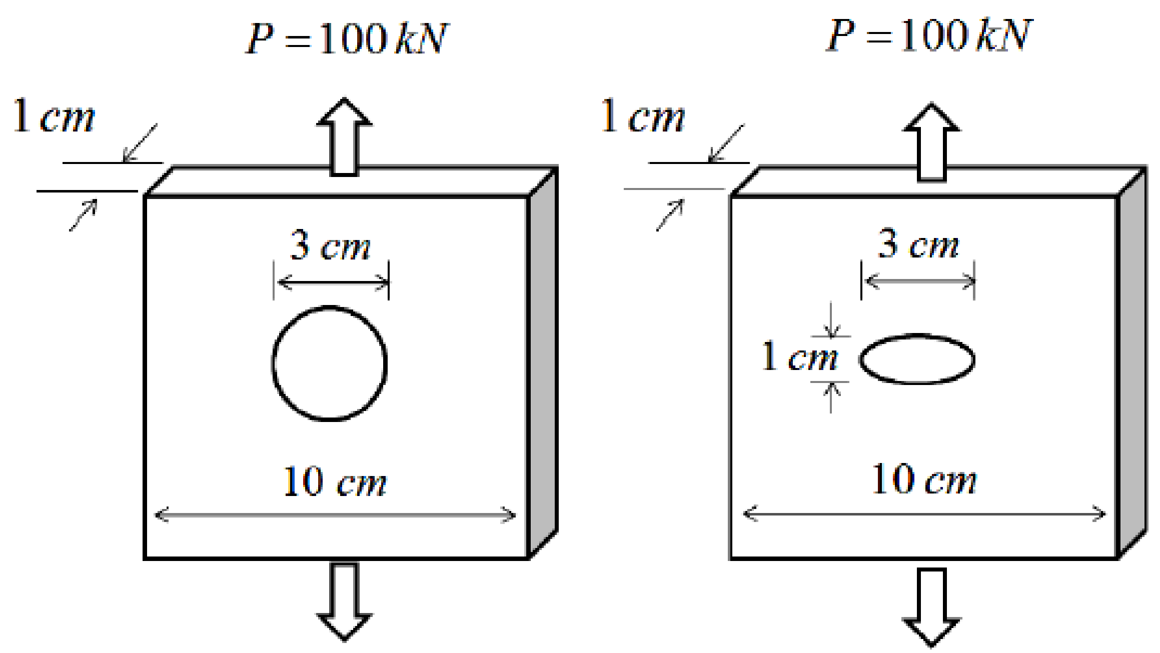
$$= 3.723 + 214.2 + \frac{(26 \times 10^3)(2.8638 \times 10^{-10}\ m)}{(347.3-200)\times 10^{-9}\ m}\ (MPa)$$

$$= 268.5\ MPa$$

PROBLEM 2

Two flat plates are being pulled in tension. The yield strength of the materials is 150 MPa. Obtain answers for both circular and elliptical holes.

A Calculate the maximum stress inside the plate.
B Will the material flow plastically?
C For which configuration is the stress higher?



PROBLEM 2 – SOLUTION

Two flat plates are being pulled in tension. The yield strength of the materials is 150 MPa. Obtain answers for both circular and elliptical holes.

A Calculate the maximum stress inside the plate.
B Will the material flow plastically?
C For which configuration is the stress higher?

(a) For the left one with the circular crack, the maximum stress can be obtained as

$$\sigma_{\max}^{circular} = \sigma\left(1+\frac{2c}{h}\right) = \frac{P}{A}\left(1+\frac{2c}{h}\right)$$

$$= \frac{100\times 10^3\ N}{10\times 10^{-4}\ m^2}\left(1+\frac{2\times 3\ cm}{3\ cm}\right) = \frac{100\times 10^3\ N}{10\times 10^{-4}\ m^2}\times 3 = 300\ MPa$$

Also, for the right one with the elliptical crack,

$$\sigma_{\max}^{elliptical} = \sigma\left(1+\frac{2c}{h}\right) = \frac{P}{A}\left(1+\frac{2c}{h}\right)$$

$$= \frac{100\times 10^3\ N}{10\times 10^{-4}\ m^2}\left(1+\frac{2\times 3\ cm}{1\ cm}\right) = \frac{100\times 10^3\ N}{10\times 10^{-4}\ m^2}\times 7 = 700\ MPa$$

(b) Yes, plastic deformation occurs for both plates near the crack tips since $\sigma_{\max}^{circular} > \sigma_y$ and $\sigma_{\max}^{elliptical} > \sigma_y$.

(c) The elliptical crack shows the higher stress state ($\sigma_{\max}^{elliptical} > \sigma_{\max}^{circular}$).

QUESTIONS ?


THE END